

The Recommendation Engine

How ShopSmart picks car parts for you, in plain English.

For: the ShopSmart team

No maths or coding needed. Read it in 15 minutes.

START HERE

The engine in one minute

The recommendation engine watches which car parts you look at and **instantly** changes the parts ShopSmart suggests to you.

You do this	The app does this, straight away
Open 2 Toyota Corolla parts	"Picked for you" fills up with Corolla parts.
Search "corolla" with max price Rs 2,000	Every suggestion is now under Rs 2,000.
Change cars and open Honda Civic parts	The suggestions follow you to the Civic.
Visit for the very first time	You see "Trending parts" until you click on something.

The 4 things to remember

1. **Fast:** each decision takes about **0 to 3 milliseconds** on our server.
2. **No sign-in needed:** it works for guests too.
3. **It explains itself:** every suggested part shows *why* it was picked.
4. **Smart split:** a fast formula ranks the parts; an AI only writes one short sentence about what you seem to need.

GLOSSARY

Words you will hear

Learn these and the rest of the guide is easy.

Word	Simple meaning
Recommendation	A part the app suggests to you.
Event	One thing you did: opened a part, or searched.
Taste profile	A short summary of what you need, e.g. "Corolla 64%, Civic 36%, shock absorbers, about Rs 5k".
Vehicle share	How much of your interest goes to one car, as a percentage.
Part type	What the part is: Brake Pads, Oil Filter, Radiator and so on (20 types).
Condition	OEM, Aftermarket, Used or Refurbished.
Recency	How recent a click is. Newer clicks count more than older ones.
Score	Points a part gets for matching your taste. Most points = shown first.
Hard budget	A strict price rule from the search price box. Nothing above it is shown.
Soft budget	A gentle price nudge from parts you viewed. Close prices get more points.
Reason badge	Small tags under a suggested part, like "Toyota Corolla" or "Under Rs 2k".
Cold start	You are brand new and the app knows nothing yet, so it shows trending parts.
Taste panel	The small "Your taste - live" box at the bottom-left of the screen.

OVERVIEW

ShopSmart has two recommenders

The live engine is new. "Similar parts" was already there and still works.

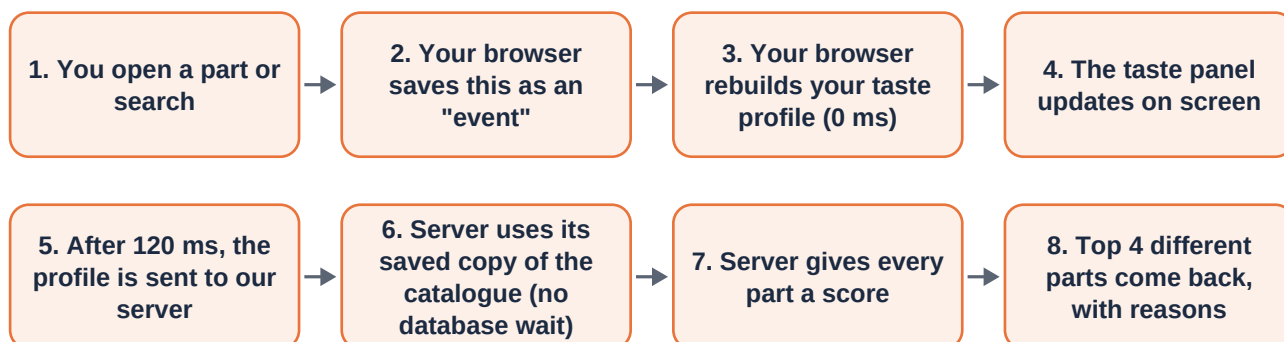
	1. Similar parts	2. Live engine ("Picked for you")
Where	Bottom of every part page	Home page, search page, part page
Looks at	The one part you are viewing	Everything you did in this visit
How it decides	AI "embeddings" (meaning-based matching) when available, otherwise same category at a similar price	A points formula: vehicle, part type, budget, condition
Updates	When the page loads	Instantly, after every click
Needs sign-in?	No	No

What changed on the home page

The home page used to have a "For you" rail that only worked for signed-in users. It is now replaced by the live **"Picked for you"** rail, which works for everyone and reacts instantly.

What happens after one click

On the live site the whole chain takes under a second, mostly internet travel time.



What the engine watches

Your action	Where	Weight	What it teaches the app
Open a part	Part page	1	The car, part type, price and condition you want
Search with a car name	Search page	2	"corolla" = Toyota Corolla, "wagon r" = Suzuki Wagon R
Search with a part name	Search page	2	"brake pads" = Brake Pads, "shocks" = Shock Absorber
Set a max / min price	Search page	-	Your budget (a strict rule)
Clear the search	Search page	-	Removes the strict budget again

The app knows 28 cars (Suzuki, Toyota, Honda, KIA, Hyundai, Daihatsu, Changan, Haval, MG, Proton) and 20 part types. If a listing doesn't say its car, the app reads it from the title. The browser keeps your last 40 events.

THE KEY IDEA

Newer clicks count more

This is why the app follows you so quickly when you switch to a different car.

Your newest click counts **100%**. The one before counts **75%**. Then **56%**, then **42%**, and so on. Older clicks fade away.

Worked example: 2 Civic parts, then 2 Corolla parts

Order	You opened	How much it counts
4th click (newest)	Toyota Corolla part	1.00
3rd click	Toyota Corolla part	0.75
2nd click	Honda Civic part	0.56
1st click (oldest)	Honda Civic part	0.42

Car	Add it up	Share of your taste
Toyota Corolla	$1.00 + 0.75 = 1.75$	64%
Honda Civic	$0.56 + 0.42 = 0.98$	36%

Same number of clicks for each car, but the Corolla wins because those clicks were **newer**. Open one more Corolla part and it rises to about **76%**. We saw exactly 64% / 36% in the live test.

What is inside your taste profile

- **Cars:** your top 5 cars with percentages.
- **Part types:** your top 5 part types (e.g. Shock Absorber 64%).
- **Condition:** the conditions you open most (OEM, Aftermarket, Used, Refurbished).
- **Budget:** a price range and a target price (see next page).
- **Already seen:** up to 30 parts you opened, so we don't suggest them again.
- **Last action:** a label like "Viewed Toyota Corolla Shock Absorber" for the panel.

How the budget works

There are two kinds of budget. The difference matters.

	Hard budget	Soft budget
Comes from	The price boxes on the search page	The prices of parts you opened
How strict	Strict: parts outside it are never shown	Gentle: close prices just get more points
Example	Under Rs 2,000	You opened parts around Rs 5,000, so about Rs 3.75k to Rs 6.25k
Panel shows	Orange badge + "hard filter"	Grey badge + "from views"

The "target price"

Scoring gives the most points to parts closest to a **target price**.

- With only views: the target is the average price of what you opened (newer clicks count more).
- With a price filter: the target is your viewed average if it fits inside the filter. If not, it is 85% of the maximum. Under Rs 2,000 means a target of Rs 1,700.

SCORING

How a part gets its points

For every part, the server adds up points. The 4 best *different* parts are shown.

What we check	Points	In plain words
Car you like	up to 10	10 x your car share. Corolla at 70% = 7 points.
Part type you viewed	up to 6	6 x that part type's share.
Price close to target	up to 4	Exactly on target = 4. Fades to 0 when 40% away.
Condition you prefer	up to 1.5	1.5 x that condition's share.
New listing (last 7 days)	0.5	Small bonus for fresh parts.
Good stock (5+ units)	0.3	Small bonus so suggestions are likely to be available.

Worked example

Your taste: **Corolla 70%, Alto 30%**. Filter: **under Rs 2,000**. Target **Rs 1,700**. Condition: **OEM 60%, Aftermarket 40%**. (All in stock, none listed this week.)

Part	Car	Price	Cond.	Stock	Total	Reason badges
Corolla Shock Absorber, Rs 1,550, Aftermarket	7.0	3.1	0.6	0.3	11.0	Toyota Corolla • Under Rs 2k
Alto Timing Chain Kit, Rs 1,800, OEM	3.0	3.4	0.9	0.3	7.6	Suzuki Alto • Under Rs 2k • OEM
City Air Filter, Rs 1,700, OEM	0	4.0	0.9	0.3	5.2	Under Rs 2k • OEM
Corolla Radiator, Rs 6,300	-	-	-	-	not shown	Above the Rs 2k filter

Result: the Corolla part comes first, the Alto part second, the City part third.

The variety rule (important for ShopSmart)

Many sellers list the same part with the same title. Without a rule, the rail could show four identical cards. So the engine **never shows the same title twice** and shows **at most 2 of the same part type**.

When does a part get a reason badge?

- **Car name**, if that car is at least 25% of your taste.
- **"Same part you viewed"**, if you opened that kind of part.
- **"Under Rs 2k"** or **"In your price range"**, if the price fits your budget.
- **"OEM"** (etc.), if that condition is at least half of your taste. At most 3 badges per part.

Which "brain" does what

We use the right tool for each job. The slow, clever part is never in the way of the fast part.

When	What decides	Speed	Why
Every click	The scoring formula (no AI)	0-3 ms	Must be instant and must never fail.
Your car or budget changes, or every 3rd part you open	GPT-4o-mini today (Claude Haiku if an Anthropic key is added)	1 to 3 s, in the background	Fast, cheap AI. One sentence needs no big brain.
You chat with the bot	The existing ShopSmart chatbot	a few seconds	Questions need a full assistant.

What the AI does

It writes **one sentence** (20 words or fewer) about what you seem to need. From the live site: *"You seem to need aftermarket shock absorbers for your Toyota Corolla, and the feed now reflects options under Rs 2k."* It shows in the taste panel as "via GPT-4o-mini".

What the AI sees, and its safety rules

It only gets the summary: car percentages, part types, budget, condition, how many parts you viewed and your last action. No name, no email, no phone number.

It must not invent listings, sellers or prices. It may only mention cars, part types, budget and condition.

If it takes longer than 4 seconds, or fails, a simple ready-made sentence is shown instead ("via Rules").

Identical profiles are cached, and the route allows 30 requests per minute per visitor.

Where you can see it

Place	Section name	Notes
Home page	"Picked for you" (or "Trending parts")	Always visible. Shows a green "Live" badge once it knows you.
Search page	"Recommended for you"	Above the results. Hidden until you have done something.
Part page	"Because you viewed this"	Never shows the part you are on. "Similar parts" sits below it.
Every page	"Your taste - live" panel	Bottom-left corner (the chatbot is bottom-right). Shrinks with the minus button.

Reading the taste panel

Part of the panel	What it means
Car bars (e.g. Toyota Corolla 64%)	Your top 3 cars and how much each counts.
Wrench line (e.g. Shock Absorber 64%)	The part type you keep opening.
Money badge (e.g. Under Rs 2k)	Your budget. Orange = strict filter, grey = from views.
Condition badge	The condition you open most (OEM, Aftermarket...).
Sparkle sentence	The AI's one-line read of what you need, and which model wrote it.
"Decided in 1ms (500ms total)"	Server time, and total time including the internet.
Vehicle-led / Part-led / Budget-led / Blended / Trending	Which signal drove the ranking.
Circle-arrow button	Reset: forget everything and start fresh.

Speed, privacy and safety

Why it is so fast

Our database is far away (about 0.4 seconds per trip). The first version asked it twice per click, about 0.8 seconds. Now the server keeps a **copy of the whole parts catalogue in memory** (one cover photo per part, about 300 KB) and refreshes it in the background every 60 seconds, so each decision needs no database trip at all.

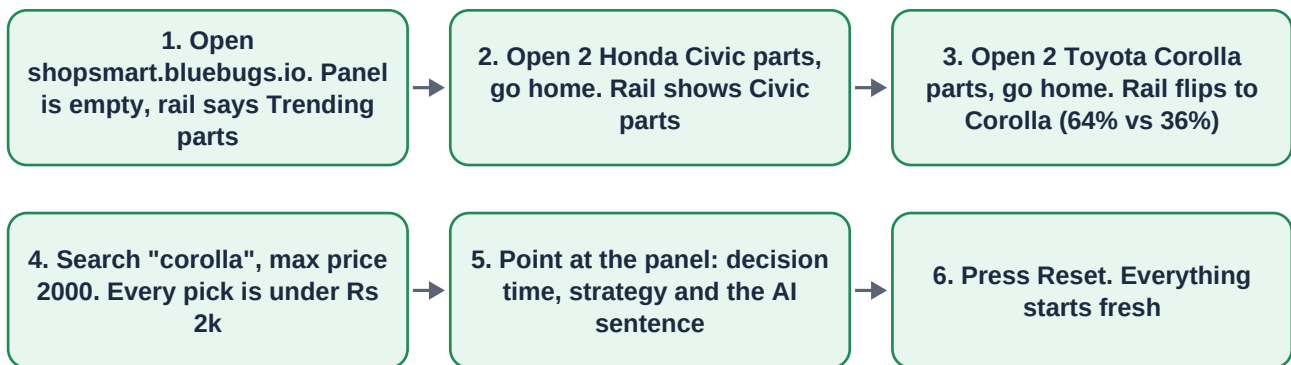
Step	Time
Updating your taste profile in the browser	0 ms (instant)
Planned wait to group fast clicks	120 ms
Server decision (scoring from memory)	about 0-3 ms (was about 0.8 s)
Total including the internet (live site)	about 0.5-0.7 s
First request after the server wakes up	about 2-4 s (loads the catalogue once)

Privacy and safety

- Your history is stored **only in your own browser**, not in our database. Reset deletes it.
- The server only receives the anonymous taste summary, never your name, email or phone number.
- The server **checks everything** it receives: at most 5 cars, 30 viewed parts, sensible price limits.
- Broken or fake requests are rejected. The AI route has a rate limit so nobody can spam it.

SHOW IT

2-minute demo script



Demo tips

- Keep the browser tab **in front**. Background tabs pause page loading, so clicks may not be recorded.
- Open the site once a minute before you present, so the server has loaded the catalogue.
- The very first suggestion after a quiet spell can take 2-4 seconds; after that the server decides instantly.

Questions people ask

Question	Answer
Why not "people who bought this also bought"?	That needs lots of past orders. Our method works from the very first click, and every pick can be explained.
Why not ask the AI on every click?	The AI takes 1 to 3 seconds and costs money each time. The formula takes about 1 ms and never breaks.
What if the AI is down?	Recommendations keep working. Only the sentence switches to a ready-made one.
Does it know which parts fit my car?	It matches the car named on each listing. A detailed fitment table (years, engines) exists but is empty for now.

BE HONEST

What it does not do yet

- **Favourites:** the engine can weigh a saved part (2.5x a view), but the heart button doesn't send it yet.
- **Exact fitment:** the compatibility table (car + years) is empty, so the car is read from the listing details or title.
- **One browser only:** your taste doesn't move between devices or follow your account.
- **No success measuring yet:** we don't yet track how often people click the suggestions.

Built and tested

Check	Result
Automatic tests	61 for the profile, scoring and request rules + 16 for the two API routes, all passing
Live test (headless browser)	Civic then Corolla flip to 64% / 36%; Rs 2k filter respected; 0 page errors
Page-error check	18 live pages (home, search, seller, 15 part pages) load with 0 React errors
Live site	shopsmart.bluebugs.io (Vercel), using the live database

For the curious: where the code lives

Part	File or address
Taste profile + car / part detection	src/lib/features/recommendations/profile.ts
Settings (weights, car + part keywords)	src/lib/features/recommendations/config.ts
Scoring formula	src/lib/features/recommendations/services/_scoring.ts
Catalogue cache + variety rule	src/lib/features/recommendations/services/live-recommendations.ts
AI sentence	src/lib/features/recommendations/services/taste-explanation.ts
Screens (rail, panel, trackers)	src/components/recommendations/
APIs	POST /api/recommendations and /api/recommendations/explanation
Technical doc	docs/LIVE_RECOMMENDATIONS.md